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SUPERSEDING
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FEDERAL SPECIFICATION

CLOTH, COTTON, FLANNEL, CANTON

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for Canton cotton flannel.

1.2 Classification. The finished cloth shall be of the following types (see 6.2):

Type I.—5.0 ounces, napped on filling side, (unbleached).

Type II. — 4.0 ounces, napped on both sides, (bleached).

Type III. — 4.0 ounces, napped on both sides, (unbleached).

Type IV.—10.0 ounces, napped on filling side, (unbleached).

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

Federal Specifications:

CCC-T-191—Textile Test Methods.

PPP-P-51 — Packaging, Packing, and Marking of Textile Fabrics (Woolens, Worsteds, Cottons, Silks and Synthetics).

Federal Standards:

Fed. Std. No. 4—Glossary of Fabric Imperfections.

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Standard:

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

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Rules and Regulations Under the Textile Fiber Products Identification Act, effective March 3, 1960.

(Copies may be obtained without charge from the Federal Trade Commission, Washington 25, D.C.)

3. REQUIREMENTS

3.1 Preproduction sample approval. When specified, before production is commenced, the contractor shall submit for approval to the contracting officer a 2-yard full-width sample of cloth he proposes to furnish (see 6.2).

3.2 Material.

3.2.1 *Yarn.* The yarn shall be single for both warp and filling and shall be made from cotton which has been carded, drawn, and spun.

3.3 Physical requirements. The finished cloth shall conform to the requirements specified in table I when tested as specified in 4.4.

3.4 Finish. Types I and IV cloth shall be scoured and napped on the filling side. Type II cloth shall be scoured, bleached, and napped on both sides. Type III cloth shall be scoured and napped on both sides. The cloth shall not contain more than 5 percent of sizing and other nonfibrous materials when tested as specified in 4.4.

3.5 Put-up. Unless otherwise specified, the cloth shall be in continuous pieces of not less than 50 yards in length and shall be put up in flat folded pieces as specified in Federal Specification PPP–P–51. When specified, the cloth shall be put up in bolts.

3.6 Fiber identification. Each piece of flannel shall be labeled, ticketed, or invoiced for fiber content in accordance with the Rules and Regulations Under The Textile Fiber Products Identification Act effective March 3, 1960.

3.7 Workmanship. The finished cloth shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Contractor's inspection responsibility. The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified,

TABLE I.—*Physical requirements*

Type	Width in inches, minimum (exclusive of selvage)	Weight per square yard (ounces)		Yarns per inch, minimum		Breaking strength (pounds) minimum		Weave
		Minimum	Maximum	Warp	Filling	Warp	Filling	
I	36	5.0	—	66	42	45	20	3 or 4 harness twill
II	36	4.0	—	38	48	24	26	Plain
III	36	4.0	5.5	38	34	20	18	Plain
IV	36	10.0	—	64	35	100	50	3/1 four harness twill

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the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction sample inspection. The preproduction sample submitted in accordance with 3.1 shall be visually inspected for appearance, nap, and finish. The sample shall be tested for physical and chemical properties in accordance with the methods specified in table III.

4.3 Examination of the end item. Except when otherwise specified by civil government agencies, examination shall be in accordance with Military Standard MIL–STD–105 except where otherwise indicated. Defects found during examination shall be classified in accordance with the list shown in 4.3.1, 4.3.2, and 4.3.3.

4.3.1 *Yard-by-yard examination.* The required yardage of each piece shall be inspected on the face side (however, the side shall be alternated on every other piece on type II and type III), and visual defects classified as listed below. The same yardage also shall be given a through lighting inspection from the back of the cloth toward a light source for holes, weak places, thin areas or heavy grease spots on the back. Single broken yarns not previously tallied on the face side of the cloth shall not be classified as defects. The defects found shall be counted regardless of their proximity to each other, except where two or more defects represent a single local condition of the cloth, in which case, only the more serious defect shall be counted. A continuous defect shall be counted as one defect for each warpwise yard or fraction thereof in which it occurs. The unit of product for this examination shall be 1 linear yard. The acceptable quality level shall be 2.5 defects per 100 units (yards). The sample size shall be based on level II of Military Standard MIL–STD–105. The lot size shall be expressed in units of 1 yard each. The number of pieces from which the sample is to be selected shall be in accordance with table II. Not more than one piece shall be selected from any one container.

Glossary Numbers of Fed. Std. No. 4	Defect
90, 91, 92	Any hole, cut or tear (including selvage).
78	Any smash.
12	Open place or crack—clearly visible.*
79	Abrasion mark—absence of nap for more than 2 inches in combined directions.
2, 9	Fine or heavy filling bar, heavy or thin place clearly visible.*
4, 10	Double or mispick—resulting in clearly visible* thick or open place.
11, 33	Broken or missing yarns—resulting in a clearly visible* open place.
3	Jerked-in-filling resulting in a clearly visible* thick place.
5, 27	Slub, slug or gout, resulting in a clearly visible* lump or thick place.
13, 14	Float—multiple 1 inch or more in combined warp and filling directions.
15, 17	Tight yarn—resulting in a waviness in the adjoining area.
74, 75, 76**	Stain, spot, or streak—clearly visible* and more than ½ inch in combined length and width directions or any uncleaned oil or grease spot regardless of size.
59, 61	Crease or wrinkle—hard embedded.

* At normal inspection distance—approximately 3 feet.

** Except defects of spot, stain or streak shall be counted regardless of cause, i.e. not restricted to rust, oil or grease.

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4.3.2 *Overall examination.* Each defect listed shall be counted not more than once in each piece examined. The unit of product for this examination shall be one piece. The sample size (number of pieces selected as sample) for this examination and the acceptance number (maximum number of defects acceptable) shall be as shown in table II.

Defects

Rancid or otherwise objectionable odor.

Overall uncleanness.

Selvage curled, folded, rolled, beaded, wavy or slack extending continuously or intermittently for more than ½ length of piece.

Uneven weaving—clearly visible.*

Not napped as specified.

Width less than specified.

* At normal inspection distance—approximately 3 feet.

TABLE II.

Lot size in yards	Sample size in pieces	Acceptance Number
Up to 1,300[1]	5	0
1,301 to 3,200	7	0
3,201 to 8,000	10	0
8,001 to 22,000	15	0
22,001 to 110,000	25	1
110,001 and up	50	1

[1] If lot contains fewer than five pieces, each piece in the lot shall be examined.

4.3.3 *Examination for length of individual pieces.* The piece shall be examined for gross length. Any gross length found to be less than the specified minimum length, or any gross length found to be more than 1 yard below the gross length marked on the piece ticket shall be considered as a defect with respect to length. The unit of product for this examination shall be one piece. The number of pieces selected as sample (sample size) for this examination and the acceptance number shall be in accordance with table II.

4.3.3.1 *Examination for total yardage in sample.* The lot shall be unacceptable if the total of the actual gross lengths of pieces in the sample is less than the total of the gross lengths marked on piece tickets. The pieces examined shall be those selected for yard-by-yard examination.

4.3.4 *Examination of preparation for delivery requirements.* An examination shall be made to determine that packaging, packing and marking requirements of section 5 of this specification are complied with. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery. The lot size shall be the number of containers in the inspection lot. The inspection level shall be L–4 and the AQL shall be 4.0 defects per hundred units.

Examine	*Defect*
Marking (exterior and interior)	Omitted, incorrect, illegible, of improper size, location, sequence, or method of application.
Materials	Any component missing. Any component damaged affecting serviceability.
Workmanship	Inadequate application of components, such as: incomplete closure of case liners, container flaps, loose strapping, inadequate stapling, bulging or distortion of containers.
Weight	Gross weight exceeds requirements.

4.4 **Testing of end item.** The methods of testing specified in Federal Specification CCC–T–191, wherever applicable and as listed in table III, shall be followed. The physical and chemical values specified in section 3 apply to the average of the determinations made on a unit of product for test purposes as specified in the applicable test methods. The unit of product shall be 1 continuous yard full width of the cloth. The lot shall be unacceptable if one or more units of product fail to meet any test requirement specified.

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The lot size shall be expressed in units of 1 yard. The sample size (number of units of product) shall be in accordance with the following:

Lot size (lot unit)	*Sample size (sample unit) Lot average and unit requirement*
Less than 800	2
801 to 22,000	3
22,001 and over	5

TABLE III.—*Test methods*

Characteristic	Requirement reference	Test method
Weave[1]	Table I	Visual
Weight	Table I	5041
Yarns per inch	Table I	5050
Breaking strength	Table I	5100
Yarn ply[1]	3.2.1	Visual
Nonfibrous material[2]	3.4	2611

[1] One determination per unit of product reported as "pass or fail".

[2] The results shall be calculated from the following formula:

$$\text{Nonfibrous content, percent} = \frac{O - S}{O} \times 100$$

5. PREPARATION FOR DELIVERY

(Federal Standard No. 102 shall be referred to for definitions and application of the various levels of packaging and packing for civil agency procurement).

5.1 Packaging. Packaging shall be level A, B, or C as specified (see 6.2 and 6.2.1).

5.1.1 *Levels A, B and C.* The cloth, put up in flat folds, shall be packaged in accordance with the applicable requirements for class 3 fabrics of Federal Specification PPP–P–51. When material in bolt form is specified by civilian government agencies, bolts shall be wrapped in waterproof or other paper as specified in the contract or order.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2 and 6.2.1).

5.2.1 *Levels A, B, or C.* The cloth shall be packed in accordance with the applicable requirements of Federal Specification PPP–P–51. When material in bolt form is specified by civil government agencies, bolts shall be packed as specified in the contract or order.

5.3 Marking. In addition to any special marking required by the contract or order, shipments shall be marked in accordance with the requirements of Federal Specification PPP–P–51.

6. NOTES

6.1 Intended use. The cloth, cotton, flannel, Canton covered by this specification is intended for use in the manufacture of padding for front interlining in coats and for the back of gloves.

6.2 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Type of cloth required (see 1.2).
(*c*) When preproduction sample approval is required (see 3.1).
(*d*) Length of pieces if other than specified (see 3.5).
(*e*) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2.1 In the preparation of contracts or orders, it must be noted that various options, choices, and alternatives, as indicated in Federal Specification PPP–P–51, may be exercised in the preparation for the delivery of cloth.

6.2.2 Level B packaging and packing shall be used by ordering agencies only when shipments are known to be made to a holding supply depot or warehouse from which partial shipments will be made to ultimate consumers.

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6.3 Standard sample. For access to standard sample, address the procuring office issuing the invitation for bids.

6.4 Transportation descriptions and minimum weight applicable to this commodity are:

Rail:

Cotton piece goods, finished, not otherwise indexed by name, in the original piece.

Carload minimum weight—Any quantity.

Motor:

Cotton piece goods, finished, not otherwise indexed, in the original piece.

Motor volume minimum weight—Any quantity.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded, by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS:
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Navy—S MC
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Copies of this specification may be purchased for

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